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Class => KRG-1-A

Experiment - 4

Approach => $f(x,y)$ => returns the no. of differing bits between x & y

~~So we need the~~ So it gives the no. of set bits in $(x \wedge y)$ or the hamming distance.

So we have to find $\Sigma f(x,y)$ over an array

So we can individually check for every bit from 0th to ~~30th bit~~ the 30th bit :-

for an ith bit => lets say out of the n numbers, i is set in 'm' numbers. So, now we can say that out of the total n^2 XORs ~~as~~ that can be obtained from any elements, $\frac{(n \times (n-m))}{2}$ XOR's will have i^{th} bit as set. So we can now directly add the ith bit's contribution to the end result

The final answer will be $2 \times \Sigma$ contribution$_i$ because the pairs are ordered.

E.g. -> 1, 3, 5 => Expected Answer = 8

001

011

101

2^0 -> set in 3 no.s

contri = $3 \times (3-3) = 0$

2^1 -> set in 1 no.

contri = $1 \times (3-1) = 2$

2^2 => set in 1 no.

contri = $1 \times (3-1) = 2$

Σ contri = 4

Ans = $4 \times 2 = 8$

Code →

```
int solve (vector <int> &a){

    vector <int> contri (32, 0);
    for(auto i: a){
        int copy = i; int ind=0;
        while (copy){
            if (copy & 1) {
                contri[ind] ++;
            }
            copy >> 1;
            ind ++;
        }
    }

    int n = nums. size;
    int res = 0;
    for (auto i: contri){
        res = (((( i * (n-i)) % mod) + res )% mod
    }

    return (res << 1) % mod ;
}
```

Time complexity => $O(n \log a_{max} + 32)$

=> $O(n \log a_{max})$

where a_{max} => largest number in array.

$(P_1 \cdot x - P_0 \cdot x) * (P_2 \cdot y - P_0 \cdot y) - ($